

1. Requirements

1.1. Functional Requirements

Req#	Requirements	Description	Comments	Rate
FR_1	Client Profile Creation	Allow the agent to create a new client record capturing: full name, phone/WhatsApp number, email, budget range, preferred property type, preferred area or neighbourhood, purchase timeframe, and lead source.	Eliminates informal note-keeping; all lead data lives in one structured place.	High
FR_2	Client Pipeline Status Tagging	Allow the agent to assign and update a status tag to every client: New Lead, Active, Long-term/Nurturing, Under Offer, Closed, or Inactive.	Makes the full sales pipeline visible and lets the agent prioritise at a glance.	High
FR_3	Active Client Overview Panel	Display a scrollable dashboard listing all active clients sorted by last-contact date, showing the date of the most recent interaction and the next scheduled follow-up for each.	Gives the agent an immediate view of who is overdue for contact.	High
FR_4	Inactivity-triggered Follow-up Reminder	Automatically send the agent a reminder notification when a client record has received no logged interaction for a user-configured period (7, 14, or 30 days).	Prevents warm leads from going cold because the agent's schedule became busy.	High
FR_5	Client Interaction Log Entry	Allow the agent to record each contact event against a client (phone call, WhatsApp message, in-person meeting) with a timestamp, channel type, and a short free-text note.	Provides a complete, dated history of every touchpoint for later reference.	High
FR_6	Long-term Nurture Check-in Scheduling	Allow the agent to mark a client as a long-term prospect and set a recurring or one-off future reminder to check in on that client.	Keeps future buyers engaged over months without relying on the agent's memory.	High
FR_7	Client Search and Filter Tool	Provide a real-time search bar and filter controls allowing the agent to locate clients by any combination of: name, status tag, budget range, location preference, and property type.	Reduces lookup time as the client list grows beyond a few dozen records.	High
FR_8	New Property Listing Entry	Allow the agent to create a property listing record with: property type, asking price, location/area, floor area, number of rooms, availability status, listing expiry date, owner name and contact, and internal notes.	Centralises all property data and removes the need to maintain separate spreadsheets.	High
FR_9	Property Availability Status Update	Allow the agent to change a listing's availability status at any time between: Available, Under Offer, Sold, and Withdrawn.	Keeps the agent's property records accurate in real time without a delay.	High
FR_10	Stale Listing Duration Alert	Automatically alert the agent when a property has remained in Available status beyond a	Prompts the agent to discuss a price reduction or	High

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		configurable threshold (30, 60, or 90 days) without a status change.	withdrawal with the owner before the listing loses credibility.	
FR_11	Asking Price Change History Log	Record every price change made to a listing, storing the previous price, new price, and date of change.	Gives the agent a documented price history to reference during owner and buyer negotiations.	Medium
FR_12	Sold and Withdrawn Listing Archiving	Allow the agent to archive or deactivate a sold or withdrawn property so it no longer appears in active searches, while preserving all its historical data.	Keeps the active listing view uncluttered without permanently deleting transaction records.	Medium
FR_13	Listing Media and Document Attachment	Allow the agent to upload and attach photos, floor plans, PDFs, and marketing materials directly to a property listing record.	Ensures all materials are accessible from a single record during client presentations.	High
FR_14	Criteria-based Property Match Query	Allow the agent to trigger a match search using a client's saved budget, preferred location, and property type to return a filtered list of currently available listings that meet all three criteria.	Eliminates manual cross-referencing of client needs against the property list.	High
FR_15	Suggested and Shown Property Linkage	Allow the agent to tag one or more listings against a client profile as 'Suggested' (recommended but not yet visited) or 'Shown' (physically presented during a viewing).	Tracks recommendation history so the agent never repeats an already-declined property.	High
FR_16	Viewing History per Client	Record and display, on each client profile, the full list of properties shown, the date of each showing, and the client's recorded reaction.	Provides a clear audit trail for follow-up conversations and future matching decisions.	High
FR_17	Property Viewing Appointment Creation	Allow the agent to create a viewing appointment by selecting a client, a property listing, a date and time, and a meeting-point location.	Replaces ad-hoc phone-based scheduling with a logged, structured appointment record.	High
FR_18	Pre-viewing Automated Reminder	Send the agent an automated push notification 24 hours before a scheduled viewing and again 2 hours before.	Ensures the agent is never caught unprepared and always shows up on time.	High
FR_19	Viewing Cancellation and Reschedule Log	Allow the agent to cancel or reschedule any viewing and record a reason and new date/time; retain the full history of changes on the appointment record.	Maintains an accurate scheduling history without overwriting earlier arrangements.	High
FR_20	Daily and Weekly Calendar Agenda	Display all upcoming appointments in a day view and a week view, showing the property, client name, time, and location for each.	Gives the agent a clear picture of scheduled commitments at any point during the week.	High

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FR_21	Post-viewing Client Feedback Capture	After a viewing, prompt the agent to log the client's reaction (Interested / Undecided / Not Interested), any verbal feedback, and a chosen follow-up action.	Captures client sentiment while it is fresh and determines the immediate next step.	High
FR_22	External Calendar Synchronisation	Sync viewing appointments to and from Google Calendar or Microsoft Outlook via two-way integration, so additions in either system are reflected in both.	Gives the agent one unified schedule without manual double-entry across tools.	Medium
FR_23	Colleague Agent Viewing Assignment	Allow the agent to record on a viewing appointment that a colleague agent will conduct it, including the colleague's name and contact.	Ensures all viewings are logged even when delegated, supporting team accountability.	Medium
FR_24	Deal Record Initialisation	Allow the agent to open a deal record by linking a buyer client, a specific property listing, and the property owner, marking the start of a formal transaction.	Creates a single record that ties every party, document, and event to one deal.	High
FR_25	Multi-stage Deal Pipeline Progression	Track each deal through the following sequential stages: Offer Initiated, Owner Notified, Negotiation in Progress, Verbal Agreement Reached, Contracts Drafted, Security Deposit Paid, Legal Review Underway, Final Sale Contract Signed, Deal Closed.	Reflects the complete Albanian real estate transaction lifecycle and prevents any stage from being skipped unrecorded.	High
FR_26	Per-stage Deal Note and Event Entry	Allow the agent to add a timestamped note or event log entry at any deal stage, recording what happened, who was involved, and what the next step is.	Maintains a chronological history of all activity so the agent can reconstruct the deal's progress at any moment.	High
FR_27	Deal Stakeholder Directory	Store and display the contact details of all parties involved in a deal on the deal record: buyer client, property owner, agent's legal representative, buyer's lawyer, and any involved colleague agents.	Provides one place to find every stakeholder without searching through separate contact lists.	High
FR_28	Transaction Document Tracker	Allow the agent to attach and label deal-related documents by type: agent-owner exclusive listing contract, agent-client showing contract, buyer-seller preliminary agreement, security deposit contract, and final sale contract.	Links all transaction paperwork to the deal record so nothing gets misplaced during a multi-week closing.	High
FR_29	At-risk Deal Flag and Reason	Allow the agent to mark a deal as At Risk and record a reason from a predefined list (financing rejected, owner withdrawal, legal dispute, client changed mind, payment delay) or free text.	Enables proactive intervention before a deal collapses and creates a record of why deals fail.	High
FR_30	Stalled Deal Stage Notification	Send the agent a notification when a deal has remained on the same stage without any logged activity for a configurable number of days (default: 7).	Prevents live deals from going silent and ensures all open transactions receive regular attention.	High

Req#	Requirements	Description	Comments	Rate
FR_31	Commission Rate Configuration per Deal	Allow the agent to select or enter the applicable commission structure for each deal: 1% (agent sourced property only), 1% (colleague sourced the client), 2% (agent sourced both), or a custom percentage.	Accurately captures the varied commission arrangements used in the Albanian market without forcing a single rate.	High
FR_32	Automatic Expected Commission Calculation	Calculate and display the expected gross commission for a deal automatically whenever the agreed sale price and commission rate are saved.	Eliminates mental arithmetic and ensures projected earnings are always immediately visible.	High
FR_33	Commission Receipt Status Tracking	Track the payment status of each deal's commission independently: Pending, Partially Paid (with amount received logged), or Fully Paid.	Gives the agent a clear and current picture of outstanding receivables across all closed and closing deals.	High
FR_34	Per-deal Expense Recording	Allow the agent to log individual expenses attributed to a specific deal, categorised as: photography, digital marketing, print marketing, fuel and travel, notary or broker fees, or other.	Enables true net profit calculation per deal beyond the gross commission figure.	High
FR_35	Earnings and Expense Summary View	Display a financial summary screen showing: total expected commissions, total commissions received, total expenses logged, and net earnings; filterable by calendar month, quarter, and year.	Provides the financial visibility the agent currently lacks and supports personal income planning.	High
FR_36	Commission-based Tax Obligation Flag	When total commissions received cross a configurable annual threshold, display a flag or alert reminding the agent to review applicable local tax obligations.	Supports compliance with Albanian tax regulations without requiring the agent to track earnings manually.	Medium
FR_37	Personal KPI Dashboard	Display a home-screen dashboard tile summary showing: total active clients, total active listings, viewings conducted this month, deals closed this month and year to date, and total commissions earned this month.	Gives the agent an instant business health check without navigating into sub-sections.	High
FR_38	Personal Sales Target Setting	Allow the agent to define monthly or quarterly targets for number of deals closed and commission earned, and display a live progress bar for each target.	Supports self-management and goal-directed focus without requiring an external spreadsheet.	High
FR_39	Exportable Business Performance Report	Allow the agent to generate and export a PDF or spreadsheet report covering a selected date range, summarising: deals opened and closed, commission earned vs. expenses, and client pipeline movement.	Enables the agent to review performance formally or share data with a business partner or accountant.	Medium
FR_40	Property Activity Insight Report	Display a list of all active listings ranked by number of viewings and flagged by days on market,	Informs owner conversations about pricing and helps	Medium

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		highlighting which properties attract the most interest and which are sitting unsold.	prioritise where to invest marketing effort.	
FR_41	Event-driven Mobile Push Notification	Send mobile push notifications for the following system events: viewing due in 24 hours, viewing due in 2 hours, client follow-up overdue, deal stage stalled, property listed too long, commission receipt overdue.	Keeps the agent informed of time-sensitive actions without requiring them to manually check the app.	High
FR_42	Notification Preference Settings	Allow the agent to enable or disable each notification type individually and configure the timing window for time-based alerts (e.g. how many days before an event to notify).	Prevents notification fatigue while ensuring the agent never misses critical alerts.	Medium
FR_43	In-app Notification History Centre	Maintain an in-app inbox listing all notifications sent to the agent in reverse chronological order, with the ability to mark items as read or dismiss them.	Provides a single place to review past alerts and ensures no notification is missed due to a dismissed lock-screen banner.	Medium
FR_44	One-tap WhatsApp and Call Launch	Allow the agent to initiate a WhatsApp chat or phone call to a client's saved number directly from the client profile screen with a single tap.	Reduces app-switching friction and keeps outreach actions tied to the client record.	High
FR_45	Cross-channel Last Contact Timestamp	Automatically update a 'Last Contacted' timestamp on a client's profile whenever the agent logs any interaction, regardless of channel (call, WhatsApp, in-person).	Feeds the inactivity-reminder logic and ensures the follow-up dashboard always reflects the true last contact date.	High
FR_46	Two-way Calendar Integration	Provide optional two-way sync of viewing appointments with Google Calendar and Microsoft Outlook, so that events created or edited in either system are reflected in REMS within 60 seconds.	Unifies the agent's schedule across tools without requiring duplicate data entry.	Medium

1.2 Non-Functional Requirements

Non-functional requirements specify the criteria that determine the overall operation of REMS rather than its individual behaviours. These include security, performance, availability, and compliance requirements. Unlike functional requirements, non-functional requirements are mandatory for ensuring a robust and usable system. Compliance with these requirements is typically evaluated with a simple "yes" or "no" response.

1.2.1 Product Requirements

1.2.1.1 Security Requirements

- Encryption of all client, property, financial, and communication data at rest and in transit (TLS 1.2 minimum).
- Authentication via password or biometric verification before accessing any part of the system.
- Role-based access control (RBAC) to restrict data sharing between agents in future multi-agent configurations.
- Client contact information must not be accessible to third parties without an explicit agent action.
- Compliance with GDPR and applicable Albanian data protection legislation.
- Secure cloud backup mechanisms to prevent data loss due to device failure, theft, or cyberattack.
- Periodic security reviews and vulnerability assessments of the production environment.

1.2.1.2 Performance Requirements

- The main dashboard must load within 2 seconds on a standard 4G mobile connection.
- Client and property search and filter queries must return results within 1 second.
- The system must support a minimum of 2,000 client records and 1,000 property listings without measurable performance degradation.
- Push notifications must be delivered to the agent's device within 60 seconds of the triggering event.

1.2.1.3 Reliability Requirements

- The system must maintain 99.5% uptime, equating to no more than 44 hours of unplanned downtime per year.
- All data entered by the agent must be auto-saved and synced to the cloud; no data should be lost due to app closure or network interruption.
- The application must function in read-only offline mode when no internet connection is available, displaying cached client and property records.
- Automatic cloud backups must run at least once every 24 hours with a recovery point objective of no more than 24 hours.

1.2.2 Organisational Requirements

1. Training and Documentation

- An in-app onboarding flow that a new agent can complete independently in under 15 minutes, requiring no prior technical training.
- Context-sensitive help tooltips accessible from within each key screen.
- A searchable FAQ and help documentation section accessible from the app's main menu.

2. Change Management

- The system must support silent over-the-air updates to the mobile application without requiring manual reinstallation.
- Critical bug fixes must be deployable to production within 48 hours of a confirmed defect report.
- Version control and rollback procedures must be maintained for all backend deployments so that a failed release can be reverted within 30 minutes.

1.2.2.1 Environmental Requirements

1. Technologies Used

- Mobile-first frontend built with a cross-platform framework (e.g. React Native or Flutter) supporting iOS and Android from a single codebase.
- RESTful or GraphQL API backend with documented endpoints.
- Cloud-hosted relational database (PostgreSQL or equivalent) with automated daily backups.
- Push notification delivery via Firebase Cloud Messaging (Android) and Apple Push Notification service (iOS).

2. Infrastructure and Hardware

- Cloud-hosted infrastructure with auto-scaling to handle peak usage without manual intervention.
- Compatibility with iOS 14 and above and Android 10 and above.
- Minimum server hardware specifications documented for any future on-premise deployment option.

1.2.2.1.3 Software Dependencies

- Full dependency list (OS, database engine, middleware, third-party SDKs) documented and version-pinned.
- Calendar integration libraries compatible with Google Calendar API v3 and Microsoft Graph API.
- WhatsApp deep-link integration using the official wa.me URI scheme.

4. Network Requirements

- Minimum recommended network bandwidth documented for optimal real-time sync performance.
- All API communication secured via HTTPS; no plain HTTP endpoints exposed.
- Firewall rules and intrusion detection configured on all cloud infrastructure.

5. Data Storage and Management

- Storage architecture designed to scale to 10,000 client records and 5,000 property listings without schema changes.
- Data retention policy defined: active records kept indefinitely; archived records retained for a minimum of 7 years.
- Personal data stored in compliance with GDPR data minimisation and purpose limitation principles.

6. Environmental Security

- All cloud infrastructure access gated behind multi-factor authentication for engineering team members.
- Audit logs of all administrative actions on the production environment retained for a minimum of 90 days.

7. Integration with External Systems

- Two-way calendar sync with Google Calendar and Microsoft Outlook.
- WhatsApp deep-link integration for direct client messaging from within the app.
- Documented public API available for future integration with Albanian property listing portals.

8. Scalability Testing Environment

- A dedicated staging environment that mirrors production, used to validate every release before deployment.
- Load and stress tests executed against the staging environment before each major release, with documented pass/fail benchmarks.

9. Regulatory Compliance Monitoring

- Quarterly internal review of compliance with GDPR and Albanian data protection legislation.
- Change log maintained for all updates affecting data handling, user authentication, or access control.

10. Usability Testing Environment

- UI and UX testing conducted with a minimum of five practising real estate agents before each major release.
- Accessibility testing for one-handed smartphone operation on both iOS and Android devices.
- Bilingual interface (Albanian and English) validated by native speakers of both languages.

11. Training Facilities

- In-app onboarding walkthrough completable without external assistance.
- Online help documentation published and accessible from a public URL, updated with each product release.

1.2.3 External Requirements

1. Integration

- Seamless integration with WhatsApp for direct client communication initiated from within the app.
- Two-way synchronisation with Google Calendar and Microsoft Outlook for appointment management.

- API-based integration capability with external property listing platforms, available for future implementation.

2. Regulatory Compliance

- Compliance with GDPR and applicable Albanian privacy law for the collection, storage, and processing of personal data.
- Explicit agent confirmation required before any client data is exported or shared externally.
- Commission and financial records maintained in a format compatible with Albanian tax reporting requirements.

1.2.3.1 Ethical Requirements

1. Fair Client Management Practices

- All client interaction logs must be timestamped automatically by the system; the agent may not backdate entries.
- Commission calculations must be deterministic and auditable, with no hidden adjustments possible.
- The matching algorithm must apply criteria consistently across all listings and must not suppress results based on any protected characteristic.

2. Responsible Use of Client Data

- Client personal and financial information may only be used for the purposes of managing the real estate transaction.
- An explicit agent action (export, share) must be required before any client contact details leave the system.
- Clients must not be profiled or scored beyond the criteria they have explicitly provided to the agent.

1.2.3.2 Legislative Requirements

1. Data Protection and Privacy

- Full compliance with GDPR and Albanian Law No. 9887 on personal data protection.
- All personal data transmitted between the mobile app and the server must be encrypted in transit.
- The system must support a right-to-erasure workflow allowing the agent to permanently delete a client record and all associated personal data on request.

2. Financial Regulations

- Commission and expense records maintained in line with Albanian tax authority requirements for self-employed individuals.
- The system must flag when cumulative annual commissions exceed a configurable threshold that may trigger a tax declaration obligation.

1.3 Domain Requirements

1. Client Lifecycle Management

- Secure and structured storage of all client profiles, interaction histories, property match records, and deal associations.
- Immutable audit trail recording every status change made to a client record, including the date and the previous value.

2. Property Listing and Matching

- Automated criteria-based matching between saved client preferences and available listings.
- Full price change history and days-on-market counter maintained per listing from creation to closure.

1.3.3 Deal and Transaction Management

- Multi-stage deal pipeline reflecting the complete Albanian real estate transaction lifecycle from first offer to final contract.
- Automatic reversion of a property listing status to Available when a deal is marked as Closed – Lost.

4. Commission and Financial Processing

- Support for multiple commission structures applicable within the Albanian real estate market.
- Automated expected commission calculation and real-time tracking of received vs. outstanding payments.
- Per-deal expense logging enabling true net profit calculation for each transaction.

5. Security and Data Protection

- Encryption of all sensitive client, property, and financial data at rest and in transit.
- Biometric and password authentication required for system access on mobile devices.

6. Scalability and Performance

- System architecture designed to scale to support a growing single-agent business and, in a future release, a multi-agent agency model.
- Performance benchmarks for search response time and dashboard load time maintained with each release.

7. Workflow and Process Automation

- Automated triggers for: follow-up reminders, pre-viewing alerts, stalled deal notifications, and long-listed property alerts.
- Automatic commission calculation on deal closure requiring no manual input from the agent.

8. Audit and Reporting Capabilities

- Exportable reports covering deal history, client pipeline movement, and financial performance across configurable date ranges.
- Detailed interaction and deal-stage change logs available for compliance and personal review.

9. Agent Experience and Accessibility

- Fully functional mobile-first interface designed for one-handed smartphone operation in the field.
- All core daily actions reachable within three taps from the home screen.
- Bilingual interface in Albanian and English selectable from the settings screen.
- Complete onboarding achievable in under 15 minutes with no external training.